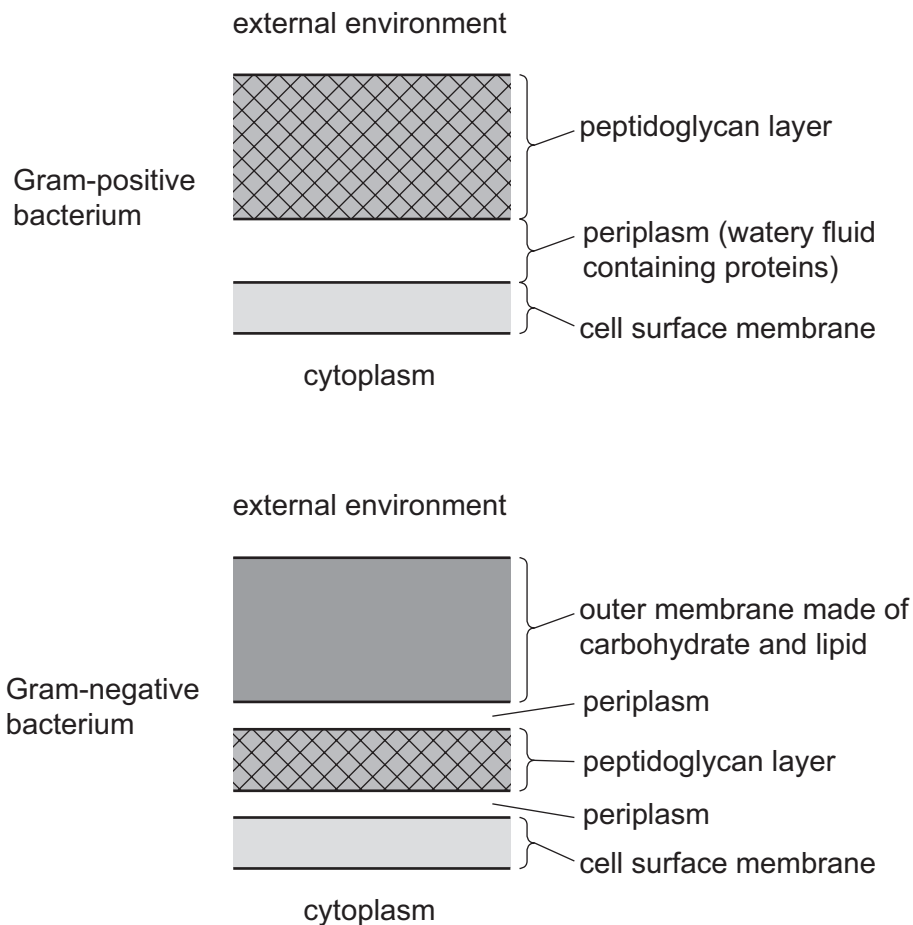


- 37** Bacteria may be classified according to differences in cell wall structure. The differences are shown by using the Gram stain.

The diagram shows part of a Gram-positive bacterium and part of a Gram-negative bacterium, drawn to the same scale.



The antibiotic penicillin kills bacteria by inhibiting the synthesis of the cell walls during bacterial cell growth.

Which type of bacteria will be killed by penicillin more easily and why?

- A** Gram-positive bacteria because the peptidoglycan layer is exposed to penicillin directly
- B** Gram-positive bacteria because it has a thinner layer surrounding the cell membrane overall
- C** Gram-negative bacteria because the thin peptidoglycan layer can be broken down faster
- D** Gram-negative bacteria because there is more periplasm available, which gives a weaker structure